Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Lesson 1: Model Selection Techniques

- Introduction to Model Selection
  Reading • 5 min
- Bayesian Model Selection
  Reading • 30 min
- Bayesian Model Selection Readings
  Reading • 1h

Lesson 2: Cross Validation

- Introduction to Cross-Validation
  Reading • 25 min
- K-Fold Cross-Validation
  Reading • 2 min
- Leave-One-Out Cross-Validation (LOOCV)
  Reading • 10 min

Lesson 3: Introduction to Optimization Techniques

- Introduction to Optimization Techniques
  Reading • 13 min
- Optimization Using Gradient Descent
  Reading • 40 min
- Gradient Descent with Momentum
  Reading • 45 min
- Stochastic Gradient Descent (SGD)
  Reading • 40 min
- **Constrained Optimization**
  Reading • 16 min
- Lagrange Multipliers
  Reading • 8 min
- Convex Optimization
  Reading • 8 min
- Linear Programming
  Reading • 15 min
- Quadratic Programming
  Reading • 10 min
- Check Your Knowledge: Optimization Techniques
  Graded Assignment • 15 min

Constrained Optimization

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In optimization problems, we often seek to minimize or maximize a function while satisfying certain constraints. These constraints can be inequalities or equalities the solution must adhere to. This section delves into the method of constrained optimization using Lagrange multipliers, which is a powerful technique for solving such problems.

Watch the following video by Khan Academy (8:43) to learn about Support Vector Machines (SVMs), a powerful machine learning algorithm used for classification tasks. This video explains the key concepts of SVMs, including decision boundaries, margins, and how they separate data into distinct classes. SVM is a practical application of the concept of constrained optimization.

Lagrange multipliers, using tangency to solve constrained optimization
Khan Academy

Watch on

The Constrained Optimization Problem

Consider the problem of minimizing a function $f(x)$ where x is a vector in R^D, subject to a set of inequality constraints given by functions $g_i(x)$ for $i = 1, \ldots, m$. Mathematically, the problem can be formulated as:

$$Minimize f(x) \quad \text{subject to } g_i(x) \leq 0 \quad \text{for all } i = 1, \ldots, m$$

This problem setup is common in many practical scenarios where the objective function $f(x)$ and the constraints $g_i(x)$ might not be convex, making the problem more challenging.

Indicator Function Approach

One naive approach to handle constraints is to introduce an indicator function $J(x)$, which imposes an infinite penalty whenever a constraint is violated:

$J(x) = f(x) + \sum_{i=1}^{m} 1\left(g_i(x)\right)$

Where $1(z)$ is an indicator function defined as:

$$1(z) = \begin{cases} 0 & \text{if } z \leq 0 \\ \infty & \text{otherwise} \end{cases}$$

While this approach theoretically enforces the constraints, it is not practical because optimizing with an infinite penalty is extremely difficult. To overcome this, we use **Lagrange multipliers.**

Go to next item ✓ Completed

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